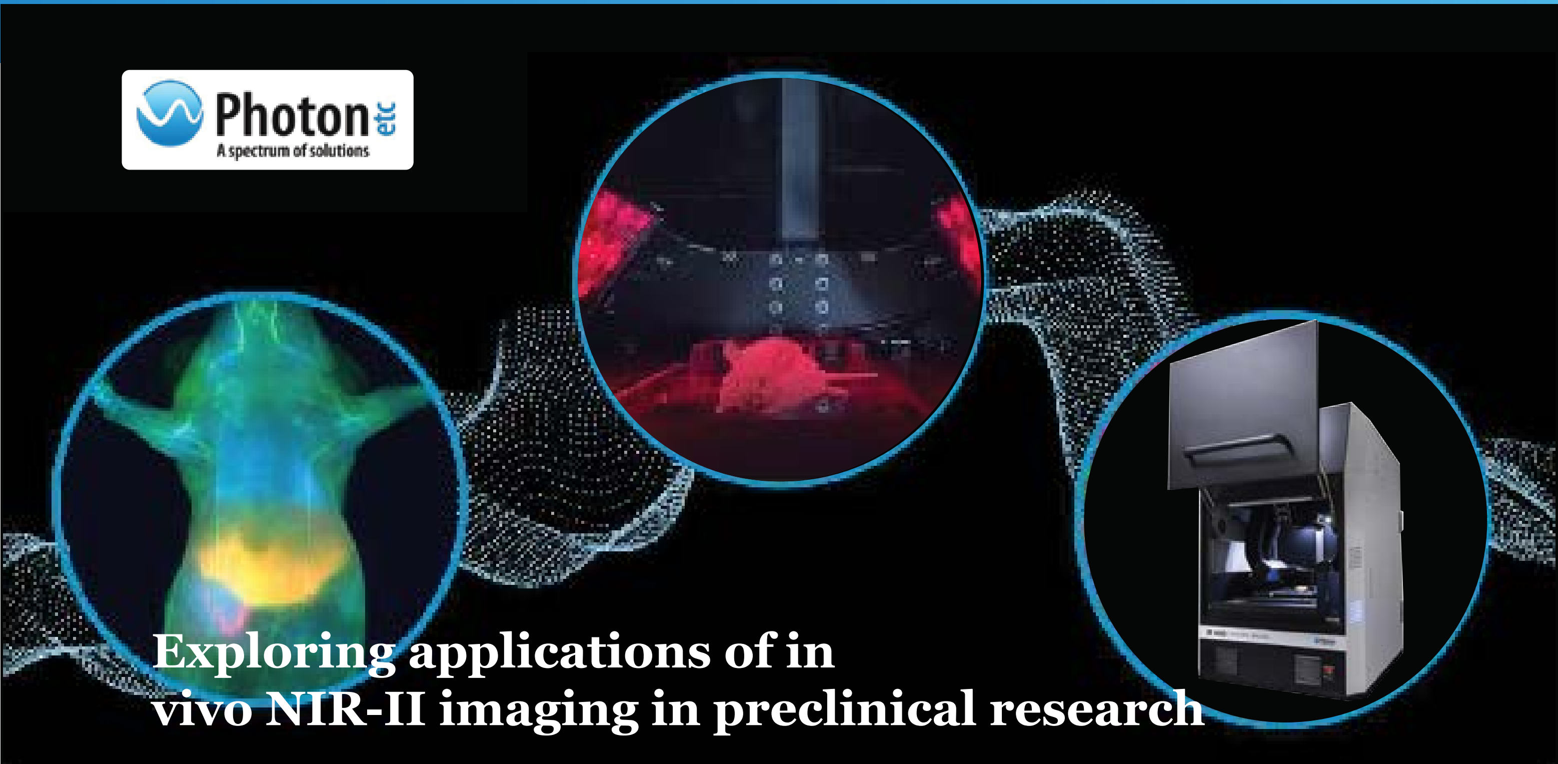

Exploring applications of in vivo NIR-II imaging in preclinical research

Disrupting preclinical research with Photon etc IR VIVO NIR II Preclinical Imager

Preclinical optical imaging suffers from the inability to localize signals due to light absorption, scattering and autofluorescence in living tissues. In vivo optical imaging can localize a signal well when it is at the surface but not when it is deep in the organism. The emergence of NIR-II offers a promising solution. NIR-II in vivo imaging is not impacted in the same way by drawbacks of light propagation in living tissues, thus enabling real-time imaging of optical probes much deeper in the organism and with much higher resolutions.

Keywords: NIR-II imaging, preclinical, near-infrared, tumour, vascular, therapeutic

Introduction

The second near-infrared spectral region (NIR-II) also referred to as shortwave infrared (SWIR) is usually defined as the wavelength range from 900 to 1700 nm.

A full understanding of why this spectral window works so well for in vivo imaging remains to be fully elucidated[1], but a few principles are clear: SWIR emitters are excited by NIR-I light which has a better penetration, and there is less autofluorescence, scattering and absorption of light in the NIR-II wavelengths of 1000 to 1700 nm, as shown in Figure 1.

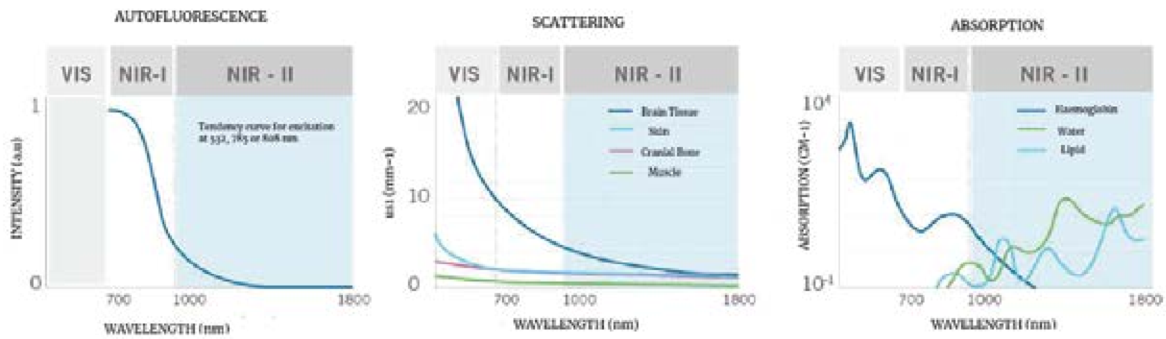

Figure 1: The weaker autofluorescence by the tissues in the NIR-II window contributes to enhancing the signal – to - background ratio. The reduced scattering and minimal absorption of emission signals by the tissues helps to improve spatial resolution and depth penetration, which can be up to 10x better.

Preclinical applications

NIR-II imaging is poised to bring us an unprecedented combination of fast, deep and high-resolution imaging at a lower cost. This will have the effect of bringing preclinical imaging techniques such as visualizing tumours, vascular anatomy, and contact-free cardiography to institutes that do not have access to more costly micro-MRI or micro-CT imaging systems. Here is a review of some of the basic preclinical applications made possible by the new modality.

Visualizing blood vessels & mapping blood flows

In vivo real-time visualization of the vascular system has great potential to improve our understanding of circulatory system-related pathologies and angiogenesis due to its ability to image vascular anatomy and blood flow. Current methods for assessing vascular structures such as micro-computed tomography (micro-CT), magnetic resonance imaging (MRI), have the advantage of unlimited penetration depth but suffer from long acquisition times and post-processing times thus complicating real-time imaging in the small animal. NIR-II imaging is also nicely suited to map arterial and venous blood flows even beyond the abilities of ultrasound at lower velocities[2]. Such blood flow mapping could help to model tumour hemodynamics and oxygenation. Moreover, these capabilities are relevant for functional imaging of activity states such as muscle motion or brain response to stimuli, which are closely linked to perfusion. Murine hindlimb models have been extensively used to look at the angiogenic effects of blood vessel growth modulation in healthy and diseased tissues such as animal models of peripheral artery disease. The high-resolution imaging of the blood vascular network will be useful for visualizing hindlimb ischemia and assessing the effectiveness of angiogenic therapies.

Assessing the effectiveness of angiogenic modulation is also important for cancer research. A solid tumour with a diameter of over 1 mm requires an adequate blood supply for growth and metastasis. Early detection and therapy research should also have an efficient and simple method to monitor tumour angiogenesis and NIR-II could represent the answer at an economical price point.

Figure 2: Data from an experiment mouse that had Ovarian tumour (SKOV-3) in the left flank following an IV injection of PbS QDots that have peak emission 1300 nm. The negative contrast in the tumour indicates a tumour barrier preventing contrast perfusion into the tumour. We also clearly see the blood vessels feeding the tumour.

Image courtesy of the Preclinical Imaging Laboratory of the National

Visualizing tumours

Early detection research will also benefit from seeing tumours earlier in the small animal. As previously mentioned, bioluminescence and fluorescence imaging have the advantage of sensitivity and detecting small changes in tumour biology, but these techniques have replaced the calliper mostly in xenograft studies where the tumour is at the surface. Visualization of tumours is possible with positive contrast caused by passive targeting (e.g. EPR effect or targeting via lymph nodes), receptor binding, or probes activated by lysosomal enzymes. More recently, HER2-amplified breast cancer BT474 cells brain parenchyma was imaged, following IV administration of Licor's IRDye800 conjugated to trastuzumab (Figure 3).

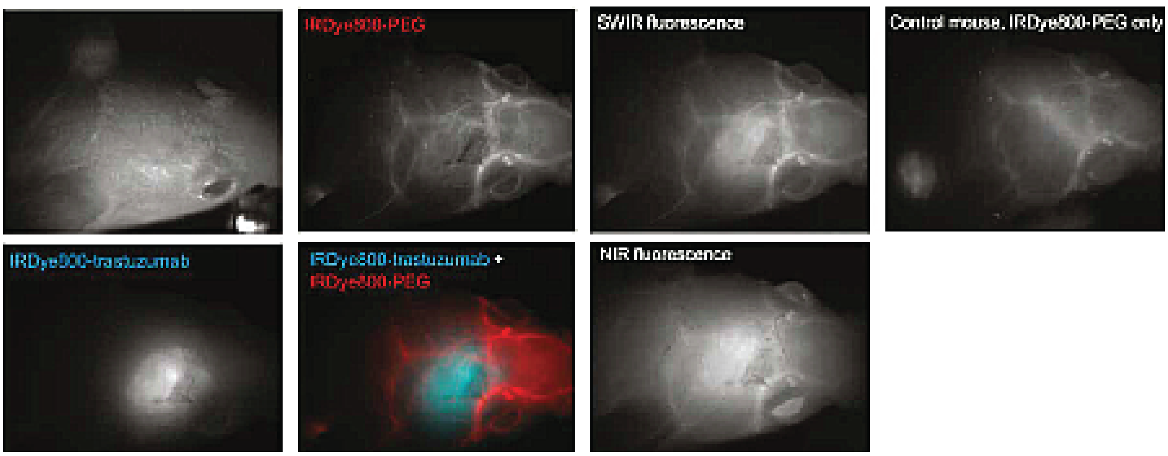

Figure 3: Targeted SWIR imaging in vivo with IRDye 800CW of a nude mouse with a brain tumour from implanted human BT474 breast cancer cells.Three days after injecting IRDye 800CW–trastuzumab conjugate, fluorescence from the labelled tumour was imaged noninvasively through skin and skull on a SWIR camera3.

In addition, a PEGylated version of the Licor dye was used to show the blood vessels surrounding the tumour[3].

Contact-free measurements of cardiography, respiratory rate & intestinal contractions

High-speed imaging made available by the fast frame rates of InGaAs cameras allows for contact-free cardiography and respiratory rate measurements in both anesthetized and awake animals. Using a blood pool contrast agent-based in micellar InAs–CdSe–CdS, short-wave infrared quantum dots with high quantum yield have generated sufficient signal-to-noise ratio for both anesthetized and awake mice showing how physiology is deeply influenced by anesthesia[4].

Such contact-free measurements of awake mice were obtained without the use of any restraining device. The monitoring of heart and respiratory rates of an anesthetized mouse was also recently observed by Photon etc. scientists in partnership with the Preclinical Imaging Laboratory of the National Research Center in Ottawa. This study also measured the mouse's intestinal contractions, values that were corroborated by scientific publications (Figure 4).

Real-time monitoring of in vivo targets for therapeutics development

NIR-II imaging has the potential to fundamentally impact the development of targeted therapeutics and drug discovery with real-time pharmacokinetic imaging of over a thousand targets simultaneously in a single mouse. The new NIR-II imaging technologies are now designed to offer the time profiling analysis for each pixel or a region of interest (ROI) in a single click, hence obtaining rapidly the real-time kinetic curves at several locations simultaneously on the mouse (Figure 5). Then, a principal component analysis (PCA) may be applied to a time series of fluorescence imaging to precisely delineate major tissues and organs.

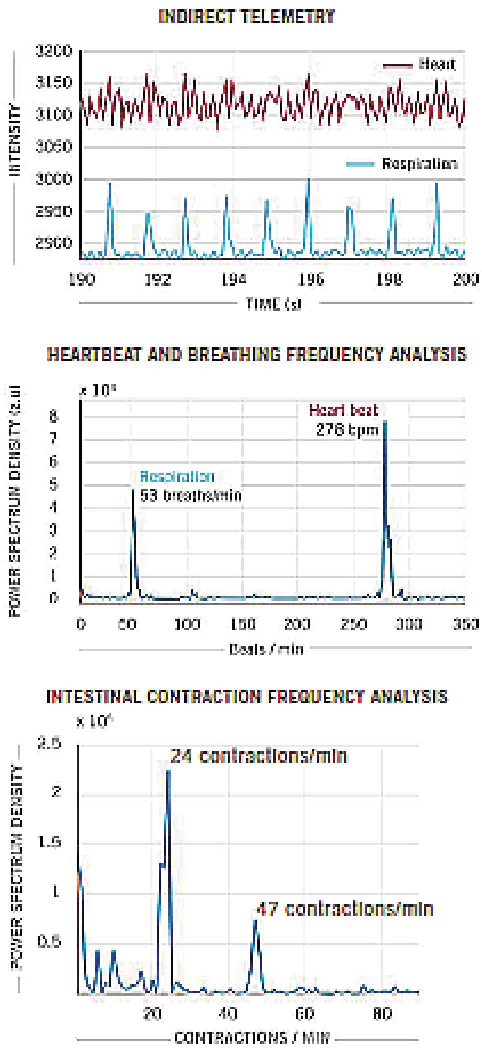

Figure 4: Real-time imaging data on heartbeat, respiratory rate and intestinal contractions obtained on Photon etc.'s IR VIVO reflect values found in the literature.

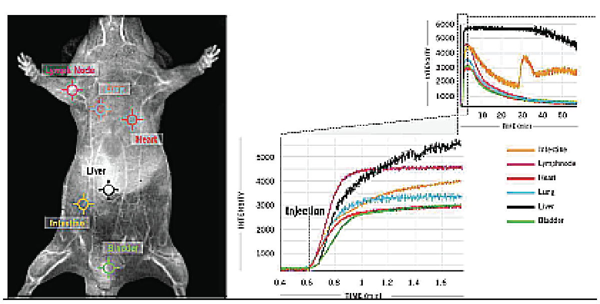

Figure 5: In this example, 6 clicks were used to obtain kinetic curves at 6 different locations - intestine, lymph nodes, heart, lung, liver, bladder) on the mouse's body for a one-hour scan following ICG injection in a male CD1 mouse.

With one click spectrum extraction, it is possible to measure the kinetics of fluorescent intensity changes allowing scientists to determine the accumulation and elimination of the probes in selected regions or organs and to compare relative signal ratios between the organs. This information helps understand how the probes are metabolized by the biological system in detail and to obtain values from hepatobiliary elimination and gastrointestinal transit rates. Such a wealth of biodistribution data from a single mouse should be of great interest to the probe development community. Historically, near-infrared imaging has been used with NIR dyes conjugated to anticancer agents for probe development. Such programs could be enhanced by simply shifting the detection technology to NIR-II and without relying on organ extraction to determine the biodistribution (Figure 6).

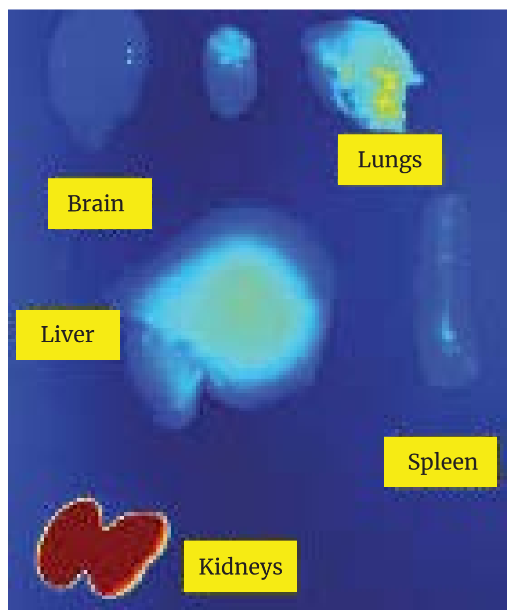

Figure 6: Studies ex vivo won't be necessary anymore to get the biodistribution data in each organ. Image courtesy of the Preclinical Imaging Laboratory of the National Research Center in Ottawa

Using the same animal for multiple PK analyses will also help to further reduce cohort sizes for preclinical studies. NIR-II could start to play a fundamental role in compound screening and allow researchers to use in vivo imaging earlier in the process of probe validation.

One-way NIR-II in vivo imaging will undoubtedly affect preclinical workflow is in study design, especially as it relates to the validation of probes in preclinical testing (Figure 7). With NIR-II in vivo imaging, it is possible to look at very small concentrations of probes flow through different ROIs of an organism and to track minute changes in real-time.

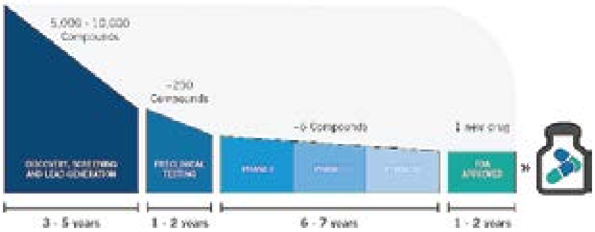

Figure 7: Therapeutics drug and development timeline.

Conclusion

NIR–II imaging has tremendous potential in preclinical optical imaging. The unique properties of near – infrared spectral region aids this technology, enabling deeper tissue penetration, higher resolution and real – time visualisation of biological processes. Photon etc's IR VIVO NIR II Preclinical Imager takes advantage of latest developments in SWIR imaging to deliver clear quantitative images.

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IR VIVO benefits from reduced light scattering, absorption and auto-fluorescence with a sensitive detection system in the near and shortwave infrared.